

PROM and HPROM Learned Reduced Models for a Parametric 2D Burgers Benchmark

Abstract

This manuscript studies learned reduced models for a two-dimensional parametric Burgers benchmark. A fixed MLSPG-sensitive basis of dimension $n_{\text{tot}} = 151$ defines the coefficient space. We first analyze a PROM-level campaign, where all intrusive online solves use the full residual and the reference coefficient trajectories are produced by a linear PROM in the same reduced space. We then repeat the baseline learned intrusive comparison at the HPROM level using empirical cubature with 2% parameter-time stratified training snapshots and direct dense SVD in the ECM preprocessing step. The PROM study compares three intrusive PROM-ANN closures, a PROM-POD-AE nonlinear-manifold solver, a direct POD-NN-ROM map, and a POD-DL-ROM latent map. The enriched PROM campaign keeps the same basis and formulation but augments the coefficient data with 36 additional PROM trajectories selected by Latin hypercube sampling. The HPROM campaign shows which conclusions remain after residual sampling is introduced: Case 1, Case 3, and PROM-POD-AE remain close to the linear HPROM reference in the baseline campaign, while Case 2 is more sensitive to the injected tail, especially for $n = 10$ and extrapolatory input. Repeating the coefficient-data enrichment at the HPROM level removes this Case-2 degradation: all learned intrusive models recover the linear HPROM accuracy scale at the four reported points. The two direct maps trained on the same linear-HPROM coefficient teacher show the same data-coverage effect: their baseline extrapolation errors are several percent, while the enriched versions return to the linear-HPROM accuracy scale without evaluating a sampled residual.

1 Objective and Scope

The first goal is deliberately restricted: given a fixed linear PROM space, can learned nonlinear closures reproduce the linear PROM coefficient dynamics and state accuracy? This isolates the learning error from any quadrature or sampling error. In the PROM sections, all intrusive results minimize the full PROM residual, and all non-intrusive results are reconstructed in the same 151-dimensional reduced basis.

The second goal is to introduce residual sampling only after the PROM-level behavior is understood. In the HPROM sections, the same learned intrusive models are evaluated with empirical cubature rules, first with nine coefficient trajectories and then with the $9 + 36$ enriched coefficient data. This makes it possible to distinguish three error sources: coefficient regression error, nonlinear feedback through the reduced residual solve, and additional residual-sampling error introduced by the HPROM approximation. The direct POD-NN-ROM and POD-DL-ROM maps are also evaluated on these HPROM-generated coefficient data; they isolate coefficient-learning error because they do not evaluate either a full or sampled residual online.

2 PROM Reference Problem

Let $\mathbf{u}(t; \boldsymbol{\mu}) \in \mathbb{R}^{N_h}$ denote the HDM state for parameters $\boldsymbol{\mu} = (\mu_1, \mu_2)$. The reduced approximation uses the fixed affine trial space

$$\mathbf{u}(t; \boldsymbol{\mu}) \approx \mathbf{u}_{\text{ref}} + \mathbf{V}_{151} \mathbf{q}(t; \boldsymbol{\mu}), \quad \mathbf{V}_{151} \in \mathbb{R}^{N_h \times 151}. \quad (1)$$

The basis $\mathbf{V}_{151}$ is the MLSPG-sensitive basis from the current PROM campaign. In contrast with a purely Euclidean snapshot POD basis, this basis was selected to be compatible with the least-squares residual solve. The basis is fixed throughout both the baseline and enriched experiments; enrichment changes only the coefficient data used to train the neural models.

At each time step, the linear PROM solves a least-squares residual problem in the full reduced space,

$$\mathbf{q}_{151}^{k, \text{lin}}(\boldsymbol{\mu}) = \arg \min_{\boldsymbol{\eta} \in \mathbb{R}^{151}} \left\| \mathbf{R}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{151} \boldsymbol{\eta}; \mathbf{u}^{k-1}, \boldsymbol{\mu}, t_k) \right\|_2^2. \quad (2)$$

The resulting trajectory $\mathbf{q}_{151}^{\text{lin}}(t; \boldsymbol{\mu})$ is used as the coefficient reference for training and for coefficient-space diagnostics. State errors are always measured against the HDM snapshots,

$$e_u(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F}{\|\mathbf{u}_{\text{HDM}}\|_F}. \quad (3)$$

Coefficient errors are measured against the linear PROM teacher,

$$e_q(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{q}_{151}^{\text{lin}} - \hat{\mathbf{q}}\|_F}{\|\mathbf{q}_{151}^{\text{lin}}\|_F}. \quad (4)$$

3 Learned Model Families

The learned models all use the same fixed coefficient space, but differ in which coordinates are solved by the PROM residual and which coordinates are predicted.

PROM-ANN Case 1. The PROM solves the first ten coordinates $\mathbf{q}_p = (q_1, \dots, q_{10})$. The secondary coordinates $\mathbf{q}_s = (q_{11}, \dots, q_{151})$ are predicted from the solved primary coordinates,

$$\mathbf{q}_s = \mathcal{N}_1(\mathbf{q}_p). \quad (5)$$

This closure is state-only. It is intrusive because $\mathbf{q}_p$ is determined by a residual minimization, but it does not receive $\boldsymbol{\mu}$ or t explicitly.

PROM-ANN Case 2 and POD-NN-ROM. A master map predicts the full coefficient vector from parameters and time,

$$\hat{\mathbf{q}}_{151} = \mathcal{N}_2(\mu_1, \mu_2, t). \quad (6)$$

The same map is used in two ways. With $n = 0$, it is the purely non-intrusive POD-NN-ROM. With $0 < n < 151$, the PROM solves $(q_1, \dots, q_n)$ by least squares and injects the tail $(\hat{q}_{n+1}, \dots, \hat{q}_{151})$ from the master map. With $n = 151$, the method becomes the linear PROM in Eq. (2).

PROM-ANN Case 3. The PROM again solves $\mathbf{q}_p = (q_1, \dots, q_{10})$, but the secondary closure receives both the solved primary coordinates and the parameter-time coordinates,

$$\mathbf{q}_s = \mathcal{N}_3(\mathbf{q}_p, \mu_1, \mu_2, t). \quad (7)$$

This is the most informative ANN closure among Cases 1–3 because it combines residual-solved coordinates with explicit parametric context.

PROM-POD-AE. An autoencoder is trained on the linear PROM coefficient trajectories. Its decoder defines a nonlinear coefficient manifold,

$$\mathbf{q}_{151} \approx \mathcal{D}(\mathbf{z}), \quad \mathbf{z} \in \mathbb{R}^{10}. \quad (8)$$

The online unknown is $\mathbf{z}$, and the PROM residual is minimized over this latent coordinate.

POD-DL-ROM. The POD-DL-ROM is non-intrusive. It learns a map from (μ_1, μ_2, t) to a latent coordinate $\mathbf{z}$, then decodes $\mathbf{z}$ to $\mathbf{q}_{151}$. This tests whether the parameter-time map is easier in a learned latent space than in all 151 coefficients.

4 Data Sets and Evaluation Points

The baseline training set is the original 3×3 parameter grid, giving nine PROM coefficient trajectories. Two additional trajectories are retained as validation data for the parameter-time master map in the enriched campaign; these validation points are not part of the four test points reported below. The baseline HPROM training records use their original 10% held-out snapshot split, while the enriched master-map record uses these two separate trajectories. The four evaluation points are

$$\begin{aligned} \boldsymbol{\mu}^{(v)} &= (4.875, 0.0225), & \boldsymbol{\mu}^{(1)} &= (4.560, 0.0190), \\ \boldsymbol{\mu}^{(2)} &= (5.190, 0.0260), & \boldsymbol{\mu}^{(3)} &= (4.000, 0.0330). \end{aligned}$$

Here $\boldsymbol{\mu}^{(v)}$ is the verification point, $\boldsymbol{\mu}^{(1)}$ and $\boldsymbol{\mu}^{(2)}$ are off-grid in-domain points, and $\boldsymbol{\mu}^{(3)}$ is an extrapolatory point.

The enriched data set keeps the same basis $\mathbf{V}_{151}$ and the same PROM solver, but adds coefficient trajectories at 36 new parameter points. Eighteen are interior Latin-hypercube samples in the original training rectangle, and 18 are exterior Latin-hypercube samples in a rectangle expanded by a 25% margin. The LHS seed is fixed to 42. No additional HDM snapshots are used to train these coefficient maps; the additional targets are generated by the linear PROM teacher.

Table 1: PROM coefficient data used for the enriched training campaign.

Quantity	Value
Baseline PROM trajectories	9
Interior LHS trajectories	18
Exterior LHS trajectories	18
Total training trajectories	45
Snapshots per trajectory	501
Training samples	22545
LHS seed	42
Margin fraction	0.25

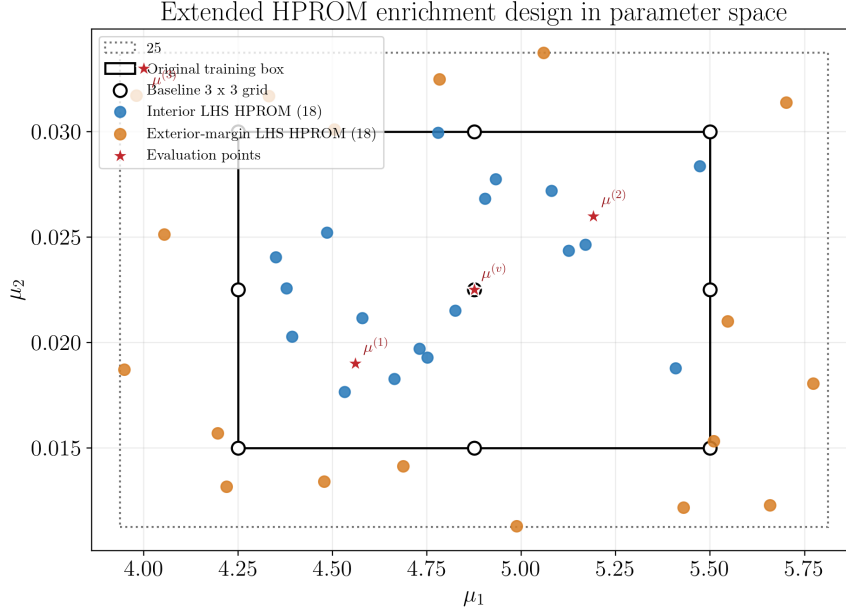


Figure 1: Parameter locations for the enriched PROM coefficient data set. The 36 additional trajectories are used only for training the learned maps; the four reported evaluation points remain unchanged.

5 Baseline PROM-Only Results

Table 2 reports the selected baseline training runs. The largest validation error occurs for the master parameter–time map because it predicts all 151 coefficients from only (μ_1, μ_2, t) . The Case 1, Case 3, and autoencoder tasks are easier because their inputs contain coefficient information.

Table 2: Baseline PROM-only Stage-3 selected networks. Errors are relative Frobenius percentages in coefficient space.

Model	learned map	parameters	train (%)	validation (%)
PROM-ANN Case 1	$(q_1, \dots, q_{10}) \mapsto (q_{11}, \dots, q_{151})$	564,621	1.130	1.251
PROM-ANN Case 3	$(q_1, \dots, q_{10}, \mu_1, \mu_2, t) \mapsto (q_{11}, \dots, q_{151})$	565,389	0.405	0.482
Master POD-NN-ROM (Case 2 tail source)	$(\mu_1, \mu_2, t) \mapsto (q_1, \dots, q_{151})$	565,399	2.601	4.391
PROM-POD-AE	$q_{1:151} \mapsto z_{1:10} \mapsto q_{1:151}$	486,817	0.106	0.127
POD-DL-ROM	$(\mu_1, \mu_2, t) \mapsto z_{1:10} \mapsto q_{1:151}$	952,747	0.810	0.935

Table 3 gives the baseline online state error. The linear PROM is not a theorem-level lower bound against the HDM, but it is the reference accuracy level associated with the fixed 151-dimensional PROM basis and teacher data. The intrusive learned PROMs are close to that reference for in-domain points. The purely non-intrusive maps degrade more strongly, especially at the extrapolatory point.

Table 3: Baseline PROM-only online errors $100 \|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F / \|\mathbf{u}_{\text{HDM}}\|_F$. The mean column averages $\mu^{(v)}, \mu^{(1)}, \mu^{(2)}$.

Model	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	mean	$\mu^{(3)}$	time (s)
Linear PROM	0.413	0.460	0.472	0.448	0.868	1289.0
PROM-ANN Case 1	0.409	0.520	0.683	0.537	1.584	222.8
PROM-ANN Case 2, $n = 10$	1.013	1.427	1.670	1.370	1.898	91.1
PROM-ANN Case 2, $n = 20$	0.999	1.328	1.505	1.278	1.658	124.1
PROM-ANN Case 3	0.405	0.521	0.576	0.501	1.392	224.2
PROM-POD-AE, $n_z = 10$	0.408	0.539	0.607	0.518	1.978	231.4
POD-NN-ROM	1.036	1.675	1.842	1.518	3.214	0.003
POD-DL-ROM, $n_z = 10$	0.489	1.398	1.463	1.116	3.728	0.006

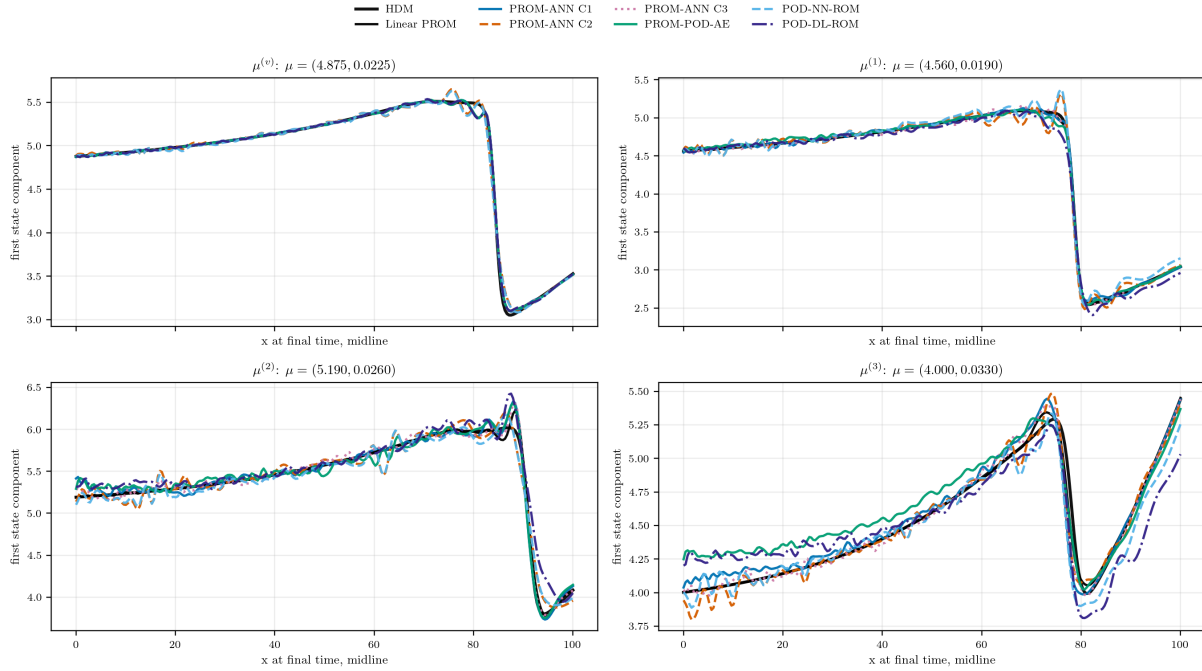


Figure 2: Baseline PROM-only final-time midline comparison against the HDM at the four evaluation points. All curves are computed from saved solver-side snapshots.

6 Case 2 Mechanism: From POD-NN-ROM to Linear PROM

Case 2 gives a useful diagnostic because it connects the direct parameter-time map to the full linear PROM. We run

$$n \in \{0, 3, 5, 10, 20, 30, 50, 100, 151\}, \quad (9)$$

where $n = 0$ is the POD-NN-ROM prediction and $n = 151$ is the linear PROM. For intermediate n , LSPG solves the first n coordinates while the master ANN supplies the remaining coordinates.

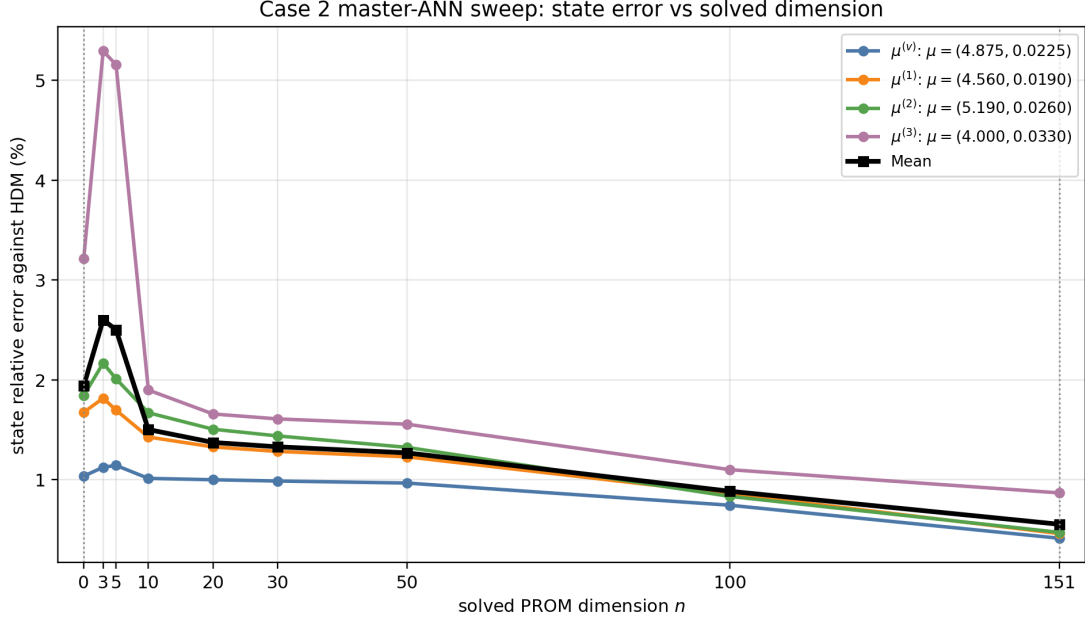


Figure 3: Baseline Case 2 state error as a function of the number of solved primary coordinates. The left endpoint is the direct POD–NN–ROM map and the right endpoint is the linear PROM.

The sweep shows that solving more primary coordinates does not automatically produce monotone improvement at all points. The injected tail still affects the residual minimization, and the benefit of increasing n depends on whether the remaining predicted tail is accurate enough for that parameter point.

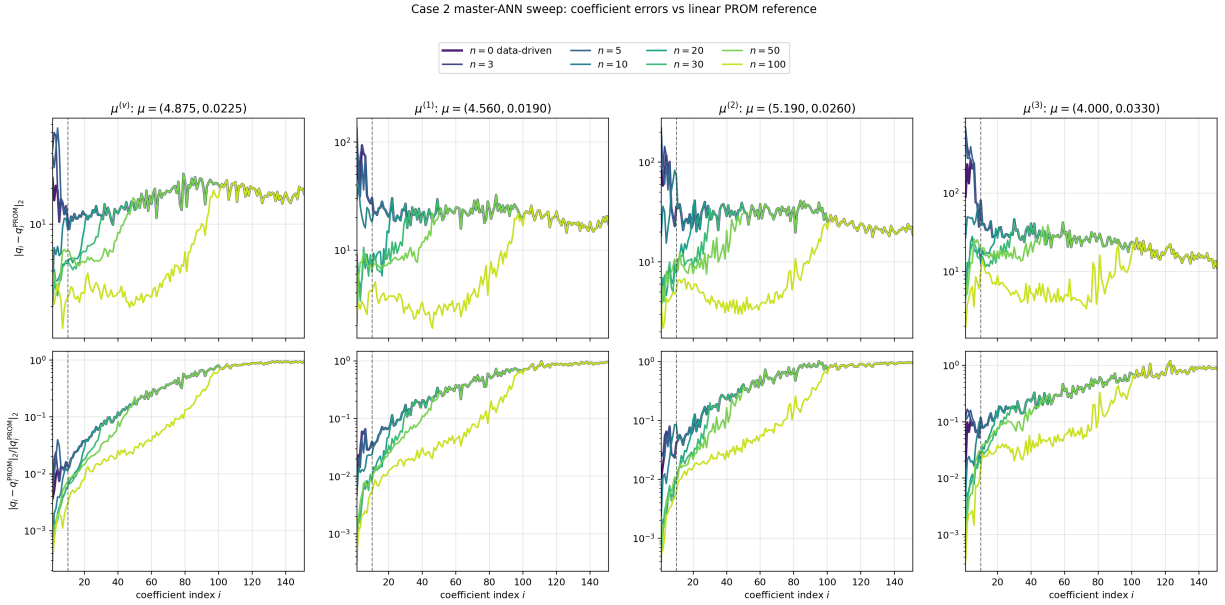


Figure 4: Baseline Case 2 coefficient errors with respect to the linear PROM coefficient trajectories for the same n -sweep. The vertical line marks coefficient 10, the primary dimension used in the main Case 2 runs.

To separate the role of the injected secondary coordinates from the LSPG solve itself, we also perturb the secondary coordinates around the linear PROM values for the fixed $n = 10$ split. This diagnostic asks how much primary-coordinate and state error is induced when the secondary tail is no longer exact.

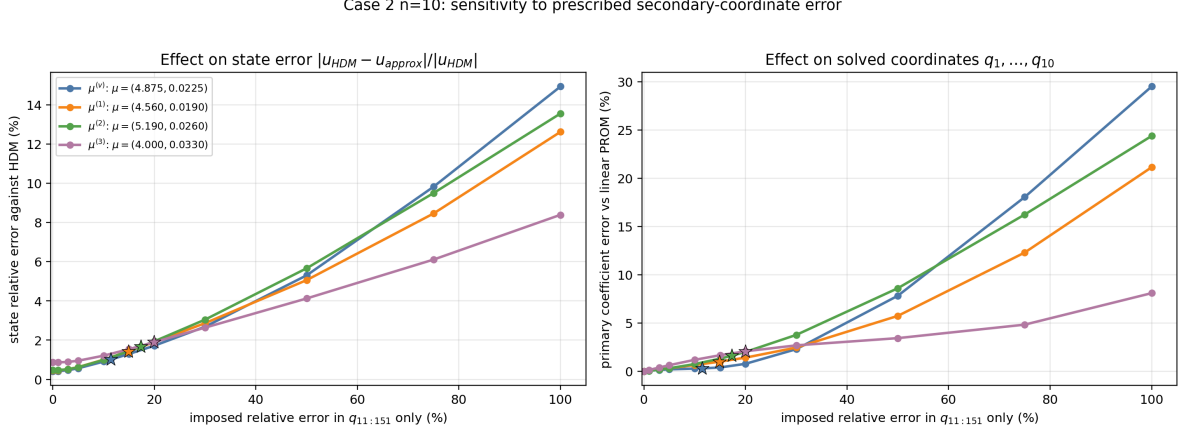


Figure 5: Baseline Case 2 secondary-coordinate perturbation diagnostic for $n = 10$. The imposed secondary-coordinate error is measured relative to the linear PROM tail; the output curves report the induced state and primary-coordinate errors.

The interpretation is that Case 2 is not only a regression problem for the tail. Tail errors can change the solved primary coordinates, so the online error reflects both tail-prediction error and residual-solve feedback.

7 Baseline Online Coefficient Diagnostics

The online state error is the final metric, but coefficient-space diagnostics explain where the learned models lose accuracy. Table 4 reports coefficient errors from the actual online outputs. Case 1, Case 2, Case 3, and PROM-POD-AE use their PROM online trajectories, while POD-NN-ROM and POD-DL-ROM use their direct non-intrusive predictions. All coefficient errors are measured against the linear PROM trajectory with $n_{\text{tot}} = 151$. For Case 1 and Case 3, the online solver did not save $\mathbf{q}(t)$ directly, so the coefficients are recovered from the saved PROM state snapshots by

$$\mathbf{q}(t) = \arg \min_{\eta \in \mathbb{R}^{151}} \|\mathbf{V}_{151}\eta - (\mathbf{u}_{\text{PROM}}(t) - \mathbf{u}_{\text{ref}})\|_2. \quad (10)$$

Table 4: Baseline online coefficient error against the linear PROM with $n_{\text{tot}} = 151$. Values are relative Frobenius percentages over all 151 coefficients and all time steps.

Point	Case 1	Case 2 $n = 10$	Case 2 $n = 20$	Case 3	POD-AE	POD-NN-ROM	POD-DL-ROM
$\mu^{(v)}$	0.295	2.440	2.401	0.164	0.140	2.499	0.814
$\mu^{(1)}$	0.593	3.361	3.123	0.856	0.773	3.924	3.129
$\mu^{(2)}$	1.297	3.906	3.512	0.788	0.929	4.249	3.225
$\mu^{(3)}$	2.923	4.257	3.662	2.396	4.394	7.444	8.639

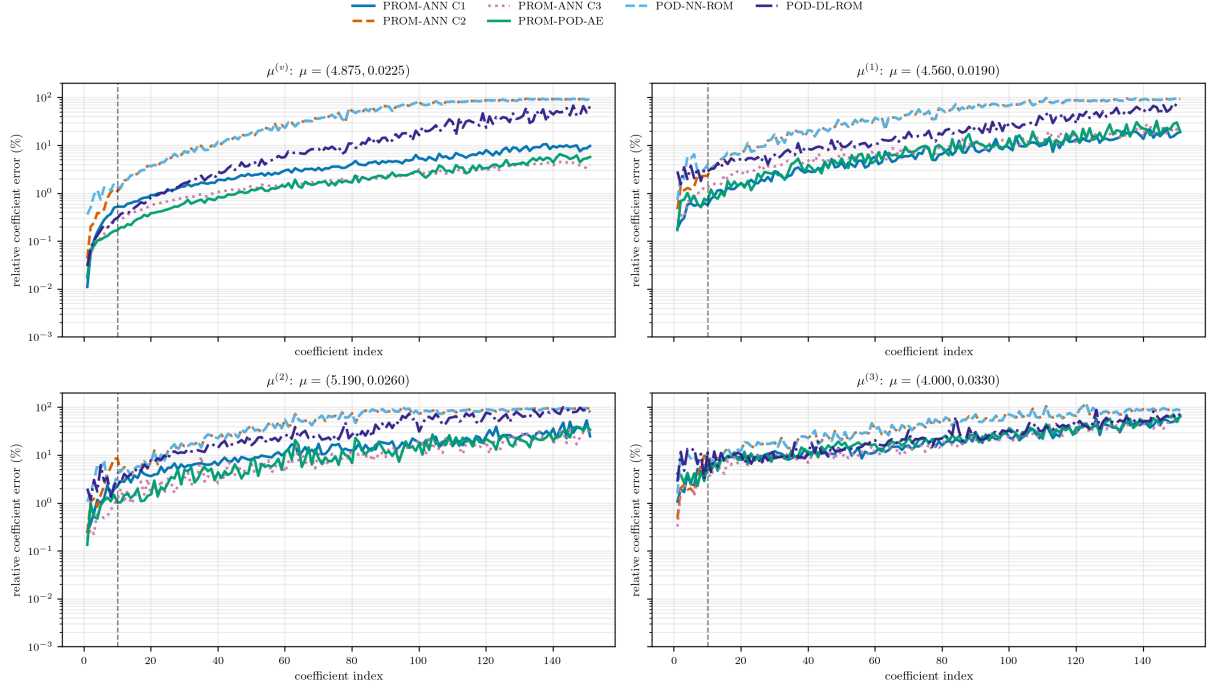


Figure 6: Baseline online per-coefficient relative errors against the $n_{\text{tot}} = 151$ linear PROM coefficient trajectory. The vertical line marks coefficient 10. The linear PROM itself is the reference and is not plotted.

The high-index coefficients often show large relative errors because their trajectory norms are small. This does not mean that each high-index coefficient dominates the state error. It does show, however, that the map $(\mu_1, \mu_2, t) \mapsto \mathbf{q}_{151}$ is data-limited in the baseline regime: the high-index tail is low energy but still difficult to interpolate from only nine parameter trajectories.

8 Effect of PROM Data Enrichment

The enriched campaign tests whether the baseline degradation is primarily an architecture issue or a data-coverage issue. The architecture is kept fixed. The basis $\mathbf{V}_{151}$, validation points, online evaluation points, and online PROM formulations are also unchanged. Only the training coefficient data are expanded from nine PROM trajectories to 45 PROM trajectories.

Table 5: Training and validation coefficient errors before and after PROM data enrichment. The validation set is kept fixed when available, so the reduction measures generalization improvement rather than a change in evaluation criterion.

Model	Base train	Base val.	Enriched train	Enriched val.	Val. reduction
Case 1	1.130	1.251	0.360	0.374	70.1%
Case 2	2.601	4.391	0.435	0.555	87.4%
Case 3	0.405	0.482	0.230	0.268	44.5%
POD-AE	0.106	0.127	0.054	0.074	42.0%
POD-DL-ROM	0.810	0.935	0.142	0.158	83.1%

The strongest gain is obtained by the master Case 2/POD–NN map: its validation error decreases from 4.391% to 0.555%. This is the clearest evidence that the previous difficulty was largely caused by the sparsity of the nine-trajectory parameter training set, not by an intrinsic impossibility of learning the map in this benchmark. The POD–DL map also improves strongly, while the already accurate autoencoder still benefits.

Table 6: Online state errors before and after PROM data enrichment. The linear PROM row is unchanged and is shown as the fixed reference accuracy level.

Model	Data	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$	Time (s)
Linear PROM	reference	0.413	0.460	0.472	0.448	0.868	1289.0
Case 1	base	0.409	0.520	0.683	0.537	1.584	222.8
Case 1	enriched	0.411	0.459	0.469	0.446	0.868	226.7
Case 2	base	1.013	1.427	1.670	1.370	1.898	91.1
Case 2	enriched	0.435	0.477	0.496	0.469	0.872	81.8
Case 3	base	0.405	0.521	0.576	0.501	1.392	224.2
Case 3	enriched	0.412	0.458	0.470	0.447	0.869	224.1
POD-AE	base	0.408	0.539	0.607	0.518	1.978	231.4
POD-AE	enriched	0.412	0.459	0.471	0.447	0.871	226.8
POD-NN-ROM	base	1.036	1.675	1.842	1.518	3.214	0.003
POD-NN-ROM	enriched	0.437	0.479	0.500	0.472	0.897	0.005
POD-DL-ROM	base	0.489	1.398	1.463	1.116	3.728	0.006
POD-DL-ROM	enriched	0.414	0.460	0.473	0.449	0.868	0.006

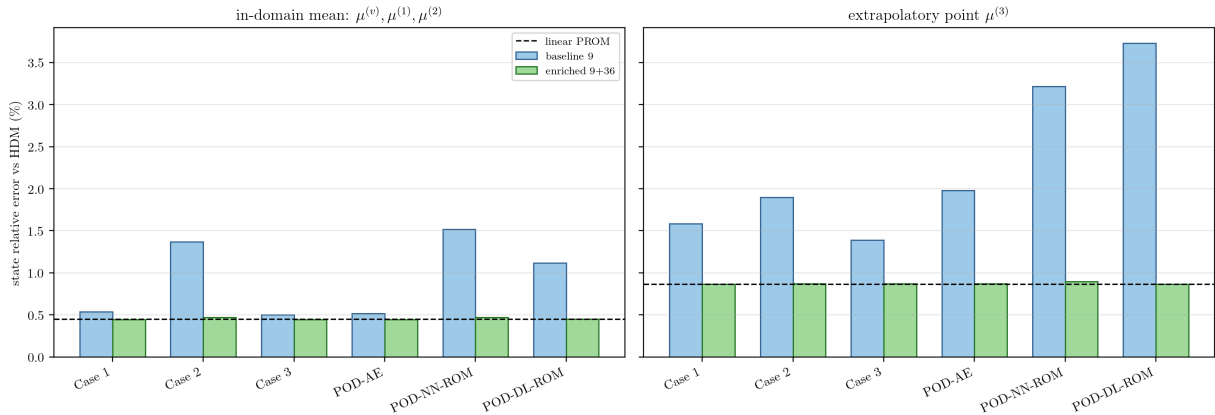


Figure 7: Effect of PROM data enrichment on online state errors. Left: mean error over the verification and two off-grid in-domain points. Right: extrapolatory point.

After enrichment, all learned models are close to the linear PROM state accuracy at the reported points. The most important qualitative change is that the non-intrusive maps no longer behave as outliers: the enriched POD–NN–ROM and POD–DL–ROM have state errors comparable to the intrusive learned PROMs. For the extrapolatory point, the enriched errors also collapse near the

fixed linear PROM level, indicating that the exterior LHS samples provide useful information in the expanded parameter region.

Table 7: Online coefficient errors against the linear PROM teacher before and after PROM data enrichment. Values are relative Frobenius percentages in coefficient space.

Model	Data	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
Case 1	base	0.295	0.593	1.297	0.729	2.923
Case 1	enriched	0.066	0.057	0.089	0.070	0.178
Case 2	base	2.440	3.361	3.906	3.236	4.257
Case 2	enriched	0.377	0.357	0.433	0.389	0.583
Case 3	base	0.164	0.856	0.788	0.603	2.396
Case 3	enriched	0.039	0.038	0.055	0.044	0.117
POD-AE	base	0.140	0.773	0.929	0.614	4.394
POD-AE	enriched	0.041	0.041	0.059	0.047	0.151
POD-NN-ROM	base	2.499	3.924	4.249	3.557	7.444
POD-NN-ROM	enriched	0.385	0.362	0.446	0.398	0.805
POD-DL-ROM	base	0.814	3.129	3.225	2.389	8.639
POD-DL-ROM	enriched	0.108	0.099	0.130	0.112	0.240

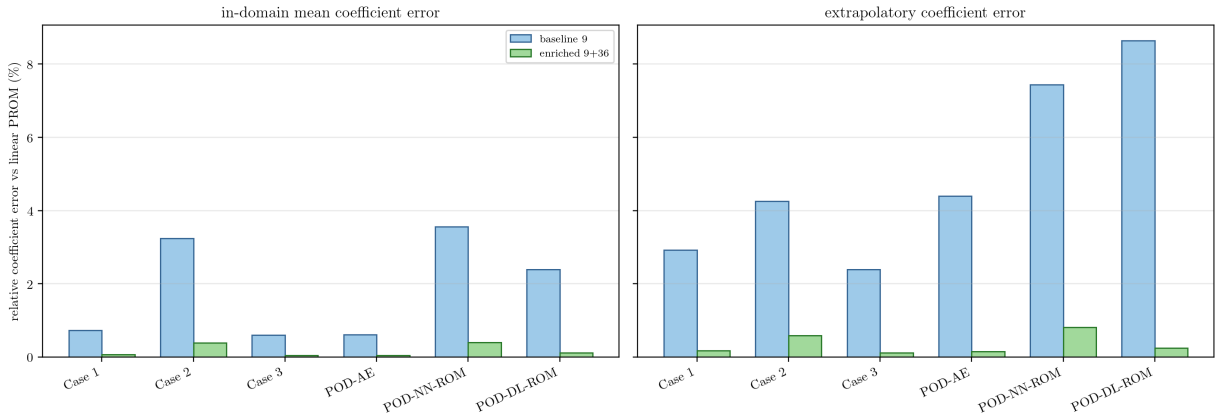


Figure 8: Effect of PROM data enrichment on online coefficient errors. The coefficient metric is stricter than the state metric because it measures the entire 151-coordinate trajectory against the linear PROM teacher.

The coefficient table confirms the state-error interpretation. The baseline POD-NN-ROM has an in-domain mean coefficient error of 3.557% and 7.444% at the extrapolatory point. After enrichment these decrease to 0.398% and 0.805%, respectively. Case 2 with $n = 10$ improves similarly because it uses the same master map for the injected tail.

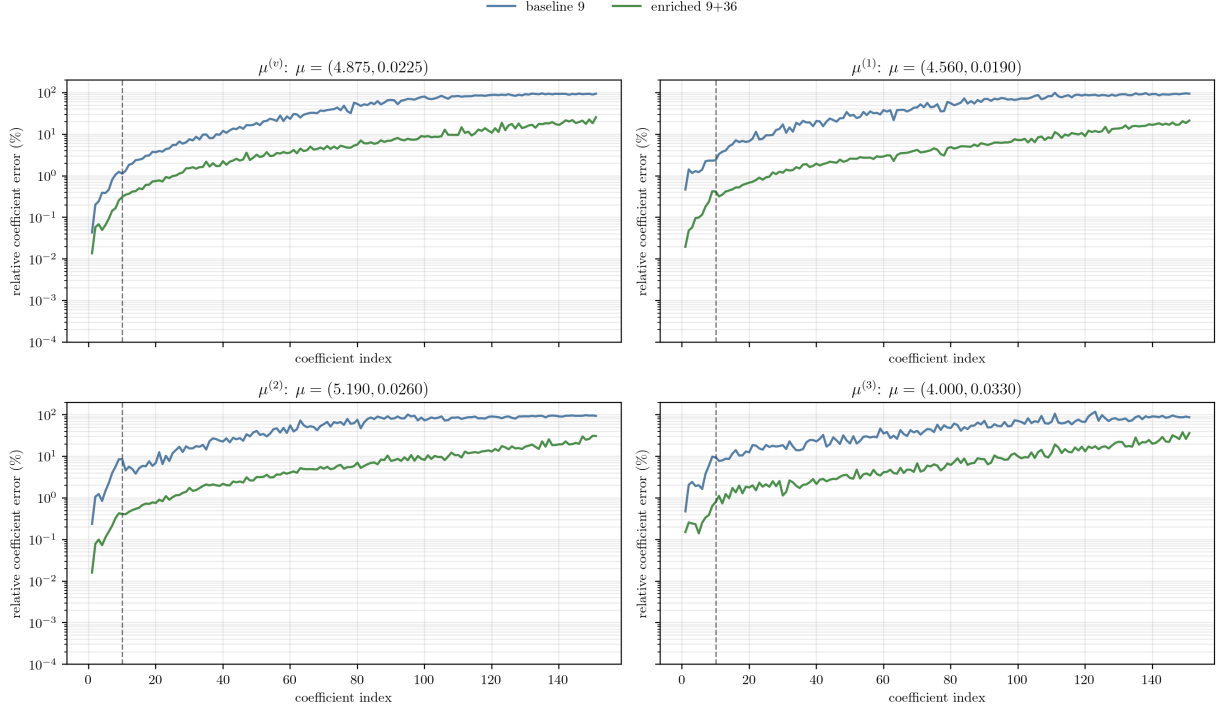


Figure 9: Case-2 per-coefficient relative error before and after PROM data enrichment for $n = 10$. The vertical line marks the separation between the ten residual-solved primary coordinates and the ANN-injected secondary tail. The linear PROM coefficient trajectory is the reference.

Figure 9 shows the mechanism relevant to the intrusive closure itself. With only nine trajectories, the ANN-injected tail has relative errors close to order one even when the global state error is moderate. The enriched training set reduces those tail errors over most coefficients, while the LSPG solve continues to recover the leading ten coordinates. This is why the Case-2 residual is far less feedback-sensitive after enrichment.

9 Baseline HPROM Results

The HPROM campaign uses the same MLSPG-sensitive basis and the same baseline nine-trajectory training set, but replaces the full residual evaluations by empirical cubature. The cubature rules are built from 2% of the available parameter-time residual snapshots using a global parameter-time stratified selection and direct dense SVD before ECM. The linear HPROM uses its own ECM rule for the 151-coordinate linear residual problem. Each learned intrusive family uses a separate rule matched to its online residual and unknown dimension. This avoids the earlier ambiguity of applying an unrelated linear rule to a nonlinear learned residual.

Table 8: Baseline HPROM cubature setup. The learned $n = 10$ rows cover Case 1, Case 2 with $n = 10$, Case 3, and PROM-POD-AE; Case 2 with $n = 20$ uses the larger $n = 20$ rule.

Family	Snapshot pct.	N_{ECSW}	SVD / sampling rule
Linear HPROM	2.0	5975	direct dense SVD, global parameter-time stratified
Learned HPROM, $n = 10$	2.0	901	direct dense SVD, global parameter-time stratified
Learned HPROM, $n = 20$	2.0	1801	direct dense SVD, global parameter-time stratified

Table 9: Baseline HPROM Stage-3 selected networks. The master map is the same parameter-time map used by Case 2; the online Case 2 solver trims this full q_{151} prediction and solves the first n coordinates.

Model	Map / network	Act.	Train e_q (%)	Val. e_q (%)	Params
HPROM-ANN Case 1	$\mathbf{q}_p \mapsto \mathbf{q}_s$; $10 \rightarrow 256 \rightarrow 512 \rightarrow 512 \rightarrow 256 \rightarrow 141$	SiLU	0.992	1.101	564621
Master map for Case 2	$(\mu_1, \mu_2, t) \mapsto \mathbf{q}_{151}$; $3 \rightarrow 256 \rightarrow 512 \rightarrow 512 \rightarrow 256 \rightarrow 151$	SiLU	0.975	1.124	565399
HPROM-ANN Case 3	$(\mathbf{q}_p, \mu_1, \mu_2, t) \mapsto \mathbf{q}_s$; $13 \rightarrow 256 \rightarrow 512 \rightarrow 512 \rightarrow 256 \rightarrow 141$	SiLU	0.491	0.577	565389
HPROM-POD-AE	z-score AE; $151 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 10 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 151$	GELU	0.098	0.119	486817

Table 10 reports the HPROM state errors. The linear HPROM is the sampled-residual analogue of the linear PROM reference. Its in-domain mean error is 0.448%, compared with 0.868% at the extrapolatory point. Case 1, Case 3, and PROM-POD-AE remain close to this reference in the three in-domain tests. Case 2 is more sensitive: with $n = 10$, the off-grid and extrapolatory errors increase substantially; increasing the solved primary dimension to $n = 20$ reduces that error, but does not make Case 2 as robust as Case 1, Case 3, or PROM-POD-AE.

Table 10: Baseline HPROM online errors $100 \|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F / \|\mathbf{u}_{\text{HDM}}\|_F$. The mean column averages $\mu^{(v)}, \mu^{(1)}, \mu^{(2)}$.

Model	N_{ECSW}	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$	Mean time (s)
Linear HPROM	5975	0.413	0.460	0.472	0.448	0.868	99.03
HPROM-ANN Case 1	901	0.423	0.575	0.702	0.567	1.547	18.57
HPROM-ANN Case 2 ($n = 10$)	901	0.512	4.516	2.895	2.641	13.098	5.81
HPROM-ANN Case 2 ($n = 20$)	1801	0.510	1.484	1.994	1.329	3.504	8.16
HPROM-ANN Case 3	901	0.415	0.514	0.592	0.507	1.416	18.90
HPROM-POD-AE	901	0.442	0.627	0.658	0.575	1.787	17.52

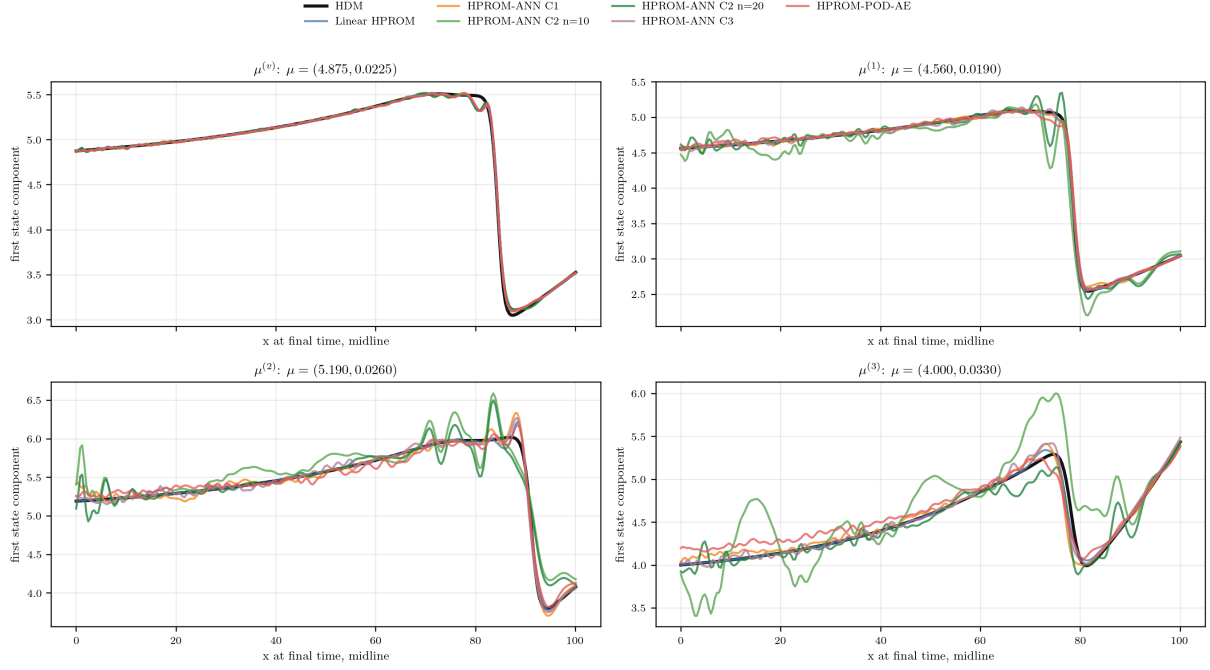


Figure 10: Baseline HPROM final-time midline comparison against the HDM at the four evaluation points. The linear HPROM curve is reconstructed from its saved coefficient trajectory; the learned HPROM curves are computed from the saved online snapshots.

The overlay in Fig. 10 shows that the large Case 2 errors are not a table artifact. For $n = 10$, the injected tail perturbs the sampled residual enough to produce visible oscillations at the off-grid and extrapolatory points. The $n = 20$ version damps much of this behavior because more of the leading coefficient subspace is recovered by the online solve.

Table 11: Baseline HPROM online coefficient error against the linear HPROM with $n_{\text{tot}} = 151$. Values are relative Frobenius percentages over all 151 coefficients and all time steps.

Model	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
Linear HPROM	0.000	0.000	0.000	0.000	0.000
HPROM-ANN Case 1	0.275	0.751	1.342	0.789	2.825
HPROM-ANN Case 2 ($n = 10$)	0.941	10.420	6.538	5.966	30.777
HPROM-ANN Case 2 ($n = 20$)	0.937	3.341	4.564	2.947	8.205
HPROM-ANN Case 3	0.240	0.734	0.907	0.627	2.544
HPROM-POD-AE	0.365	0.936	1.160	0.820	3.950

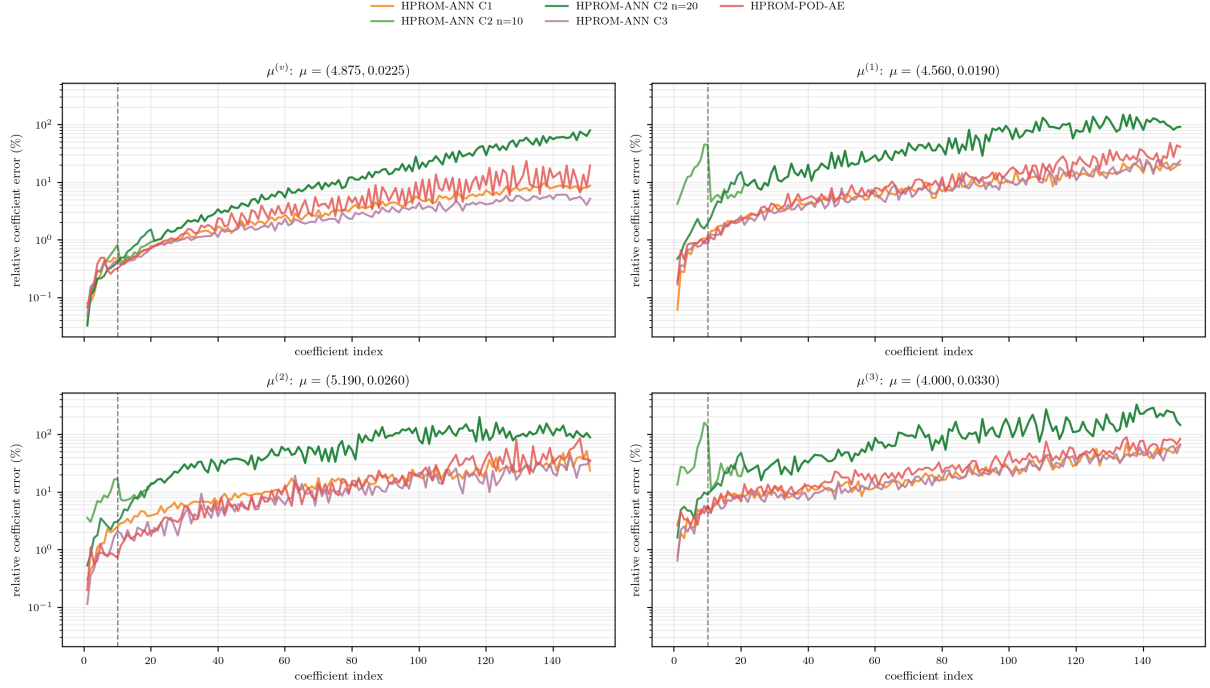


Figure 11: Baseline HPROM online per-coefficient relative errors against the $n_{\text{tot}} = 151$ linear HPROM coefficient trajectory. The vertical line marks coefficient 10. The linear HPROM itself is the reference and is not plotted.

The HPROM coefficient diagnostics mirror the PROM-level mechanism. The high-index tail is the difficult part of the learned map, and Case 2 is the most exposed to this difficulty because its tail is injected directly into the online residual. The $n = 20$ Case 2 run improves both state and coefficient errors relative to $n = 10$, but the remaining tail error is still large for the extrapolatory point. Case 3 is the most accurate learned HPROM in coefficient space because it combines residual-solved primary coordinates with explicit (μ_1, μ_2, t) information in the closure.

9.1 Case-2 HPROM Diagnostic Sweep

The Case-2 mechanism can be examined separately from the production comparison by varying the number of residual-solved primary coordinates. The two endpoints are the direct parameter-time map at $n = 0$ and the 151-coordinate linear HPROM at $n = 151$. For every intermediate value $n \in \{3, 5, 10, 20, 30, 50\}$, we rebuilt a matched Case-2 ECSW rule from the same nine parameter trajectories using 2% global parameter-time stratified residual snapshots, seed 42, and direct dense SVD. Thus, the intermediate rows are a diagnostic of the coupled choice of primary dimension and matched cubature rule, not a continuation with one fixed sampled residual.

Table 12: Case-2 HPROM diagnostic state errors $100 \|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F / \|\mathbf{u}_{\text{HDM}}\|_F$. The intermediate rows use newly built, matched 2% ECSW rules; $n = 0$ and $n = 151$ are the existing direct-map and linear-HPROM references. The mean averages $\mu^{(v)}, \mu^{(1)}, \mu^{(2)}$.

Solved n	N_{ECSW}	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
0	—	0.508	1.834	2.479	1.607	7.401
3	271	0.534	2.855	2.495	1.961	16.816
5	451	0.518	3.872	2.302	2.230	21.154
10	901	0.512	3.840	2.731	2.361	9.471
20	1801	0.509	1.481	1.972	1.321	3.280
30	2700	0.506	1.405	1.858	1.256	3.334
50	4260	0.510	1.297	1.652	1.153	2.865
151	5975	0.413	0.460	0.472	0.448	0.868

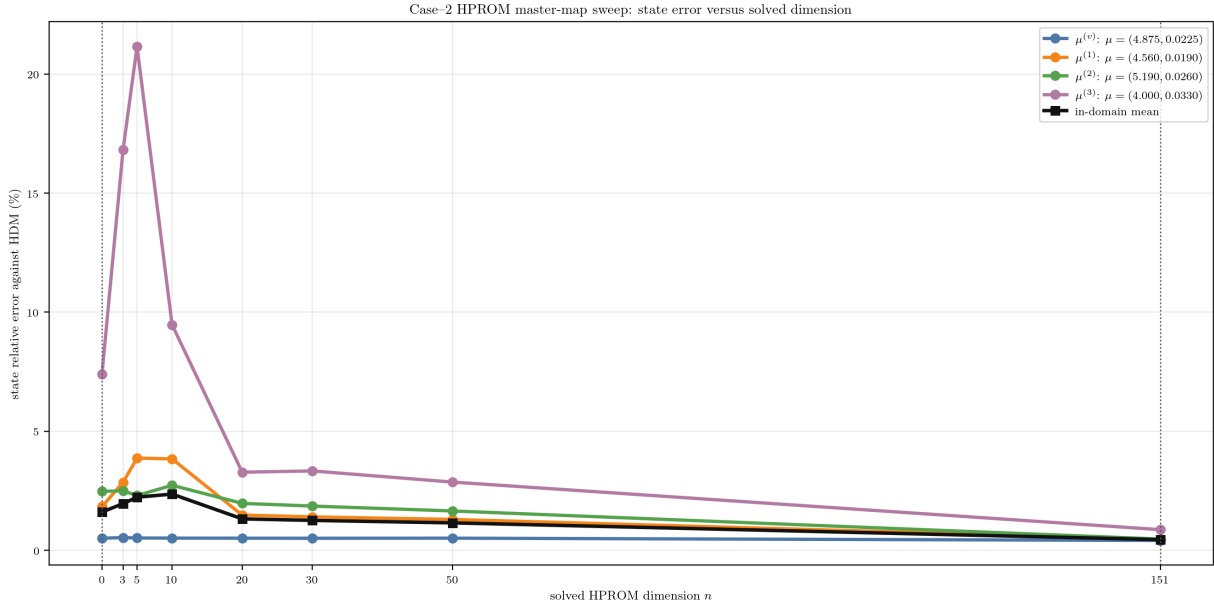


Figure 12: Case-2 HPROM state error as the number n of residual-solved primary coordinates is varied. Each intermediate point uses its matched 2% ECSW rule. The dashed horizontal levels are the 151-coordinate linear-HPROM errors.

The small- n behavior is non-monotone. The direct map has an in-domain mean error of 1.607% and an extrapolatory error of 7.401%. Solving three or five coordinates does not reliably improve this result: the corresponding in-domain/extrapolatory pairs are 1.961%/16.816% and 2.230%/21.154%. Once $n \geq 20$, the matched HPROM errors decrease to 1.321%/3.280% at $n = 20$, 1.256%/3.334% at $n = 30$, and 1.153%/2.865% at $n = 50$. These values remain above the

linear-HPROM reference (0.448%/0.868%), but show that the degradation is not an unavoidable consequence of solving only a subset of coordinates.

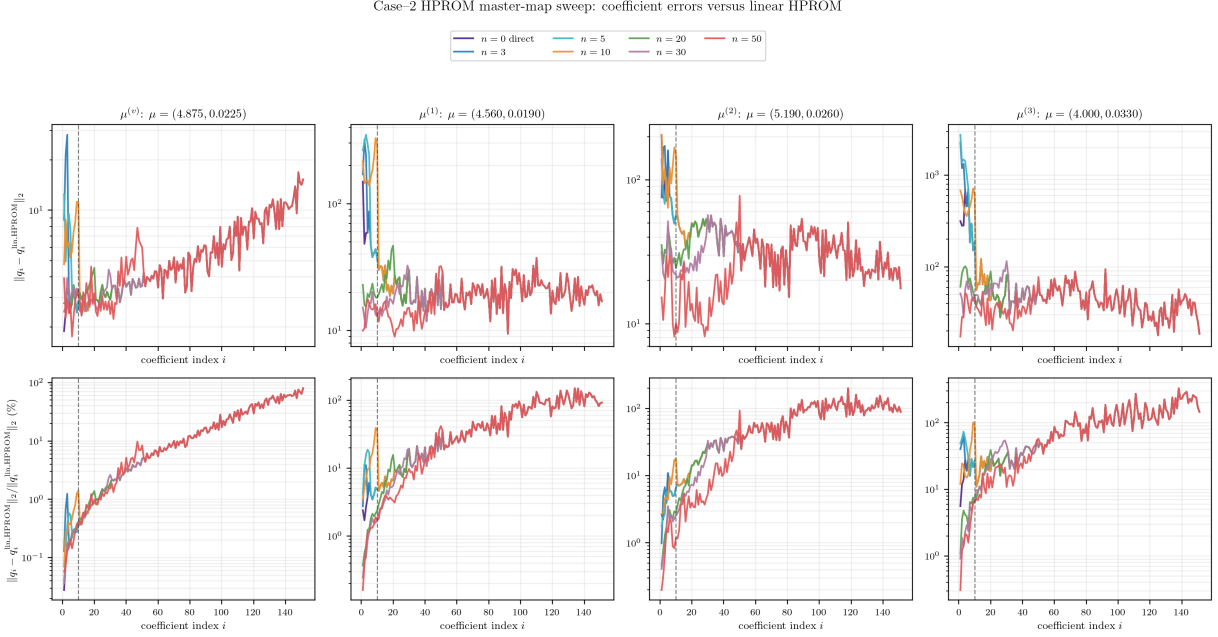


Figure 13: Case-2 HPROM coefficient errors for the matched-rule diagnostic sweep. Top: absolute time-trajectory error. Bottom: relative error with respect to the linear HPROM. The dashed vertical line separates the first ten coordinates. The $n = 151$ linear HPROM is the zero-error reference and is omitted.

Figure 13 shows that the residual solve chiefly reduces errors inside the solved leading block. The secondary tail remains the output of the same master map, so increasing n moves the boundary of the correction without making the learned tail exact. The resulting state trend is therefore a coupled consequence of a larger solved subspace and a different sampled residual, rather than a simple monotone interpolation between the two endpoints.

9.2 Fixed-Rule Tail-Error Sensitivity

To isolate tail feedback from the changing-rule effect, we fixed the production Case-2 $n = 10$ ECSW rule ($N_e = 901$) and imposed a controlled secondary-coordinate perturbation $\mathbf{q}_s^{\text{used}} = \mathbf{q}_s^{\text{lin,HPROM}} + \alpha(\hat{\mathbf{q}}_s - \mathbf{q}_s^{\text{lin,HPROM}})$. Here $\hat{\mathbf{q}}_s$ is the master-ANN tail and α scales its actual error direction from zero to 50% of the linear-HPROM tail norm. The primary coordinates are still obtained by the same sampled LSPG solve. Stars in Fig. 14 mark the actual master-ANN tail error at each test point.

Table 13: Fixed-rule Case-2 HPROM tail diagnostic for $n = 10$; all error columns are percentages. The master-ANN tail error is measured against the linear HPROM. The zero-tail row is the constrained ten-coordinate sampled solve with the exact linear-HPROM tail; it is not the 151-coordinate linear-HPROM solution.

Point	ANN tail error	e_u at exact tail	e_u at ANN tail	e_{q_p} at ANN tail	$e_{q_{151}}$ at ANN tail
$\mu^{(v)}$	4.353	0.452	0.512	0.184	0.941
$\mu^{(1)}$	14.695	0.564	4.516	10.164	10.420
$\mu^{(2)}$	21.740	0.475	2.895	4.865	6.538
$\mu^{(3)}$	38.799	1.077	13.098	30.445	30.777

Case-2 HPROM, $n = 10$: sensitivity to prescribed secondary-coordinate error

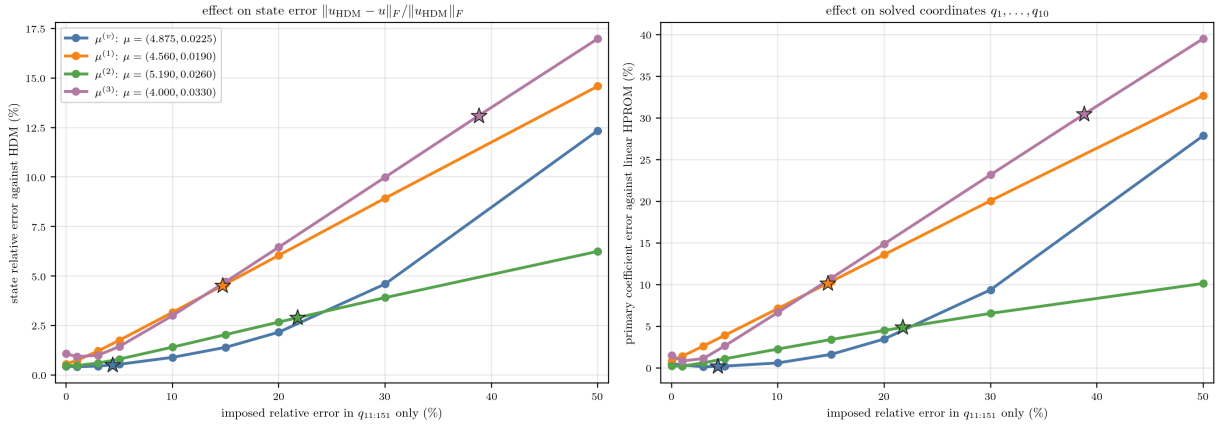


Figure 14: Fixed-rule sensitivity of Case-2 HPROM with $n = 10$ to secondary-coordinate error. Left: state error against the HDM. Right: error in the ten solved primary coordinates against the linear HPROM. Stars identify the actual master-ANN tail errors.

The fixed-rule result establishes direct residual feedback. Even an exact linear-HPROM tail leaves a nonzero zero-tail baseline because a constrained ten-coordinate sampled solve is not the full 151-coordinate linear-HPROM problem. Additional tail error then changes the solved primary coordinates: at off-grid point $\mu^{(1)}$, a 14.695% master-ANN tail error produces a 10.164% primary-coordinate error and a 4.516% state error; at the extrapolatory point, a 38.799% tail error produces 30.445% and 13.098%, respectively. Hence Case 2 is not a passive reconstruction map: its learned tail enters the online residual and can shift the coordinates that are nominally solved intrusively.

10 Effect of HPROM Coefficient-Data Enrichment

The HPROM enrichment campaign uses the same 36 Latin-hypercube parameter locations as the PROM study: 18 interior points and 18 points in the 25% expanded exterior rectangle, with seed 42. The fixed basis, the 2% parameter-time sampling ratio, direct dense SVD, and the nine baseline parameter locations used to construct each nonlinear ECM rule are retained. The nonlinear residual maps do change with the retrained neural networks, so a rule is rebuilt for each enriched learned model, but no new HDM trajectory or residual-training parameter location is introduced. Consequently, this experiment isolates the effect of enlarging the linear-HPROM coefficient teacher

data from 9 to 45 trajectories.

Table 14: Reported coefficient-space training and validation errors for the HPROM learned maps before and after coefficient-data enrichment. Values are relative Frobenius percentages. The baseline records use their original internal held-out snapshot split; the enriched master-map record uses the two separate validation trajectories, so the validation columns are useful diagnostics but are not a strictly paired model-selection protocol.

Model	Base train e_q (%)	Base val. e_q (%)	Enriched train e_q (%)	Enriched val. e_q (%)
HPROM-ANN Case 1	0.992	1.101	0.358	0.373
Master map for Case 2	0.975	1.124	0.404	0.489
HPROM-ANN Case 3	0.491	0.577	0.202	0.234
HPROM-POD-AE	0.098	0.119	0.049	0.066

The training diagnostics improve for every intrinsic closure. For example, the reported Case 1 validation error decreases from 1.101% to 0.373%, Case 3 decreases from 0.577% to 0.234%, and the master parameter–time map decreases from 1.124% to 0.489%. The online comparisons below are the decisive test, because all four points are kept out of both training campaigns.

Table 15: Baseline and enriched HPROM state errors $100 \|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F / \|\mathbf{u}_{\text{HDM}}\|_F$. Each mean averages $\mu^{(v)}, \mu^{(1)}, \mu^{(2)}$. The linear HPROM is identical in both columns because its basis and ECM rule are fixed.

Model	Baseline (9 trajectories)					Enriched (9+36 trajectories)				
	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
Linear HPROM	0.413	0.460	0.472	0.448	0.868	0.413	0.460	0.472	0.448	0.868
HPROM-ANN Case 1	0.423	0.575	0.702	0.567	1.547	0.417	0.504	0.486	0.469	0.854
HPROM-ANN Case 2 ($n = 10$)	0.512	4.516	2.895	2.641	13.098	0.434	0.471	0.500	0.468	0.921
HPROM-ANN Case 2 ($n = 20$)	0.510	1.484	1.994	1.329	3.504	0.424	0.467	0.485	0.459	0.911
HPROM-ANN Case 3	0.415	0.514	0.592	0.507	1.416	0.435	0.517	0.476	0.476	0.848
HPROM-POD-AE	0.442	0.627	0.658	0.575	1.787	0.510	0.664	0.498	0.557	0.919

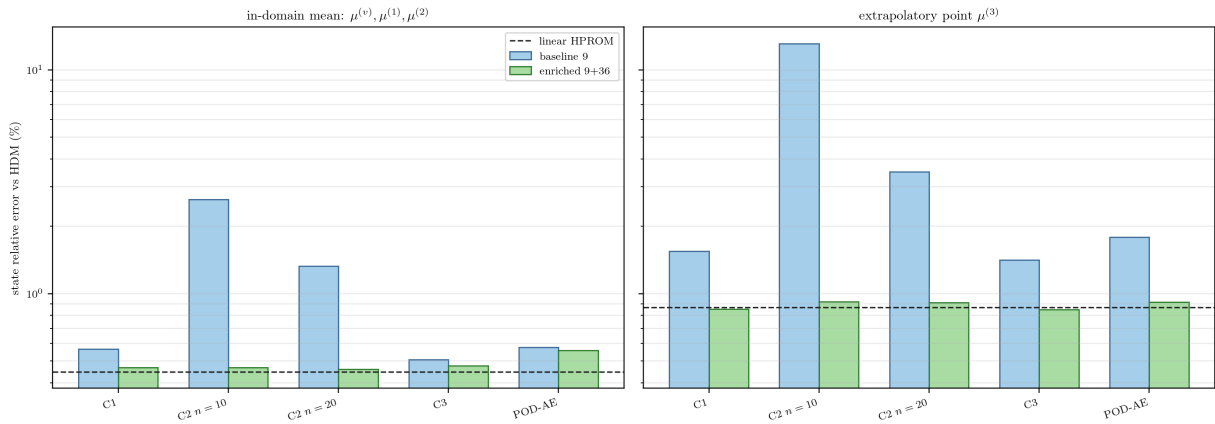


Figure 15: Effect of HPROM coefficient-data enrichment on state error. The dashed level is the fixed linear HPROM reference; logarithmic axes make both the baseline Case–2 degradation and the enriched recovery visible.

The baseline sensitivity of Case 2 disappears after enrichment. At $n = 10$, its in-domain mean state error falls from 2.641% to 0.468%, and the extrapolatory error falls from 13.098% to 0.921%. The latter is close to the 0.868% linear HPROM reference. Solving 20 coordinates remains slightly better in coefficient space, but no longer provides the qualitative rescue that was needed with only nine trajectories. Case 1 and Case 3 also remain close to the fixed linear HPROM level, while PROM-POD-AE stays accurate but retains a modest gap at the first off-grid point.

Table 16: Baseline and enriched coefficient errors against the corresponding fixed $n_{\text{tot}} = 151$ linear HPROM coefficient trajectory. Values are relative Frobenius percentages over all coefficients and all time steps.

Model	Baseline (9 trajectories)					Enriched (9+36 trajectories)				
	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
Linear HPROM	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
HPROM-ANN Case 1	0.275	0.751	1.342	0.789	2.825	0.170	0.305	0.442	0.306	0.584
HPROM-ANN Case 2 ($n = 10$)	0.941	10.420	6.538	5.966	30.777	0.469	0.377	0.546	0.464	0.876
HPROM-ANN Case 2 ($n = 20$)	0.937	3.341	4.564	2.947	8.205	0.352	0.311	0.412	0.358	0.706
HPROM-ANN Case 3	0.240	0.734	0.907	0.627	2.544	0.202	0.366	0.269	0.279	0.550
HPROM-POD-AE	0.365	0.936	1.160	0.820	3.950	0.599	0.919	0.449	0.656	1.177

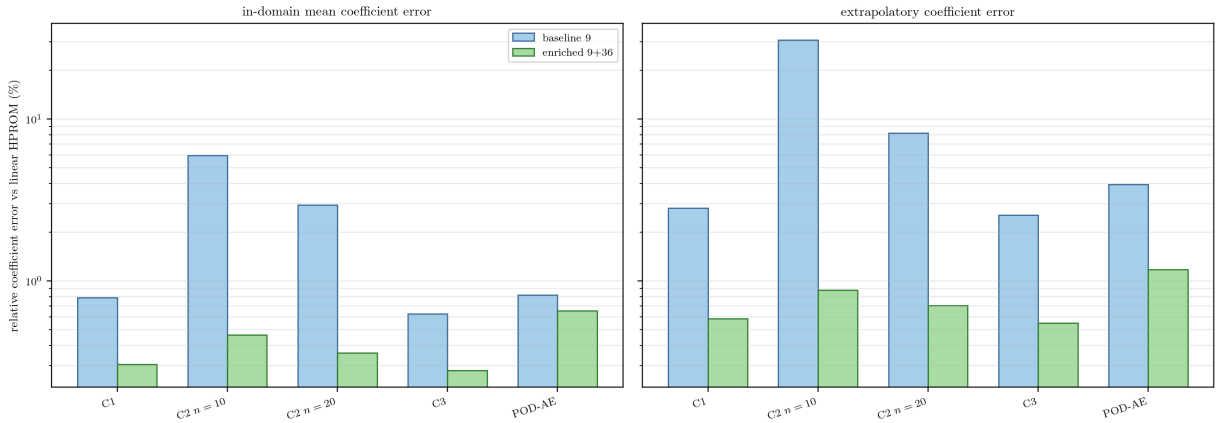


Figure 16: Effect of HPROM coefficient-data enrichment on global coefficient error. Enrichment reduces the Case-2 tail error by more than one order of magnitude at the extrapolatory point, while retaining sub-percent global coefficient errors for the other intrusive closures.

Coefficient space confirms that this is not merely cancellation in the state norm. The Case-2 $n = 10$ in-domain mean coefficient error decreases from 5.966% to 0.464%, and its extrapolatory coefficient error decreases from 30.777% to 0.876%. The enriched Case-2 $n = 20$ values are 0.358% and 0.706%, respectively. The linear HPROM reference is unchanged; the observed improvement therefore comes from coefficient coverage rather than a different linear basis or a more accurate linear cubature rule.

Direct maps trained on linear-HPROM data. POD-NN-ROM and POD-DL-ROM do not evaluate a residual and therefore are not HPROM solvers; they have no ECM approximation error. We nevertheless evaluate both maps in the baseline and enriched campaigns because this isolates the

quality of the linear-HPROM coefficient teacher and the coverage of the parameter–time regression problem.

Table 17: Baseline and enriched online accuracy of the two direct maps trained on linear-HPROM coefficient data. Each coefficient error uses the corresponding linear HPROM trajectory as teacher. This is not an HPROM-versus-ROM comparison, because neither direct map evaluates an empirical-cubature residual.

Model	coefficient data	State error vs HDM (%)				Coefficient error vs linear HPROM (%)					
		$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
POD–NN–ROM	baseline (9)	0.508	1.834	2.479	1.607	7.401	0.929	4.192	5.642	3.588	17.433
POD–NN–ROM	enriched (9+36)	0.423	0.466	0.487	0.458	0.926	0.341	0.300	0.396	0.345	0.752
POD–DL–ROM	baseline (9)	0.451	1.483	1.498	1.144	4.753	0.616	3.334	3.291	2.413	11.101
POD–DL–ROM	enriched (9+36)	0.414	0.460	0.473	0.449	0.869	0.096	0.093	0.116	0.102	0.243

The baseline direct maps expose the same coverage limitation as Case 2. The POD–NN–ROM in-domain mean state error falls from 1.607% to 0.458% after enrichment, and its extrapolatory error falls from 7.401% to 0.926%. The corresponding POD–DL–ROM values fall from 1.144% to 0.449% in-domain and from 4.753% to 0.869% at extrapolation. Thus, the improvement is due entirely to coefficient-data coverage, not to a sampled residual solve. Figure 17 shows that the coefficient errors decrease in parallel, ruling out an interpretation based only on cancellation in the state norm.

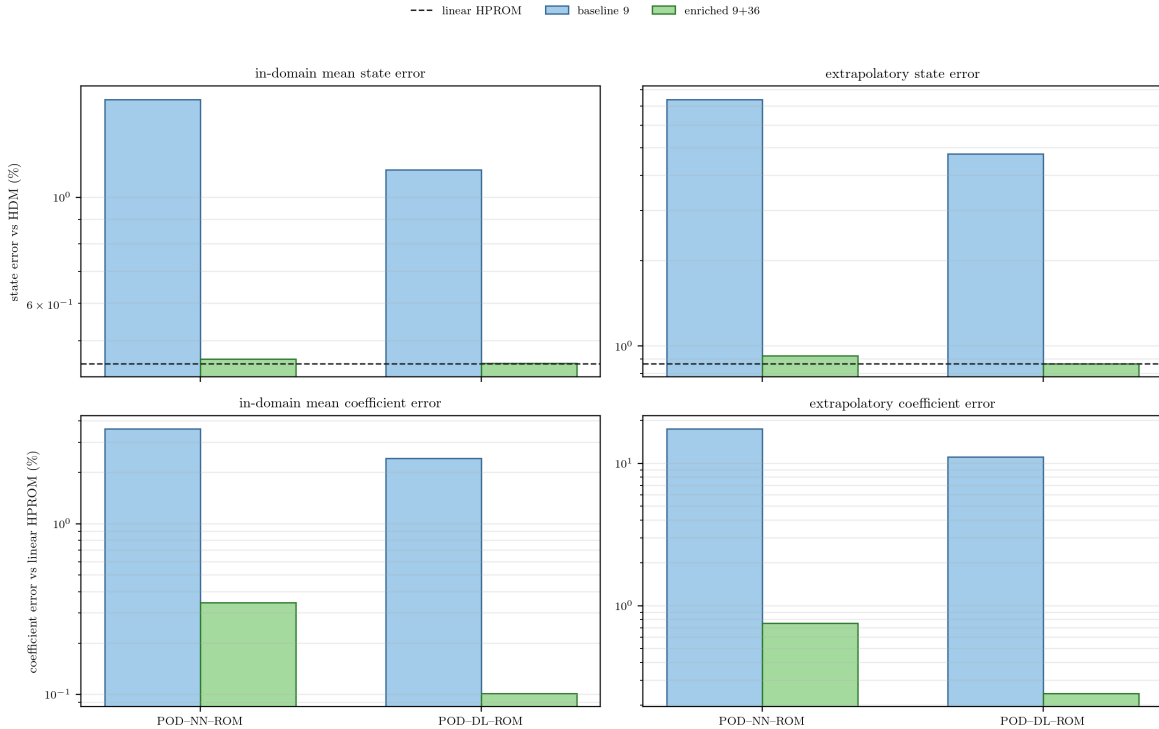


Figure 17: Effect of linear-HPROM coefficient-data enrichment on the two direct non-intrusive maps. The top row shows state error against the HDM; the bottom row shows coefficient error against the corresponding linear HPROM teacher. The dashed state-error level is the linear HPROM reference.

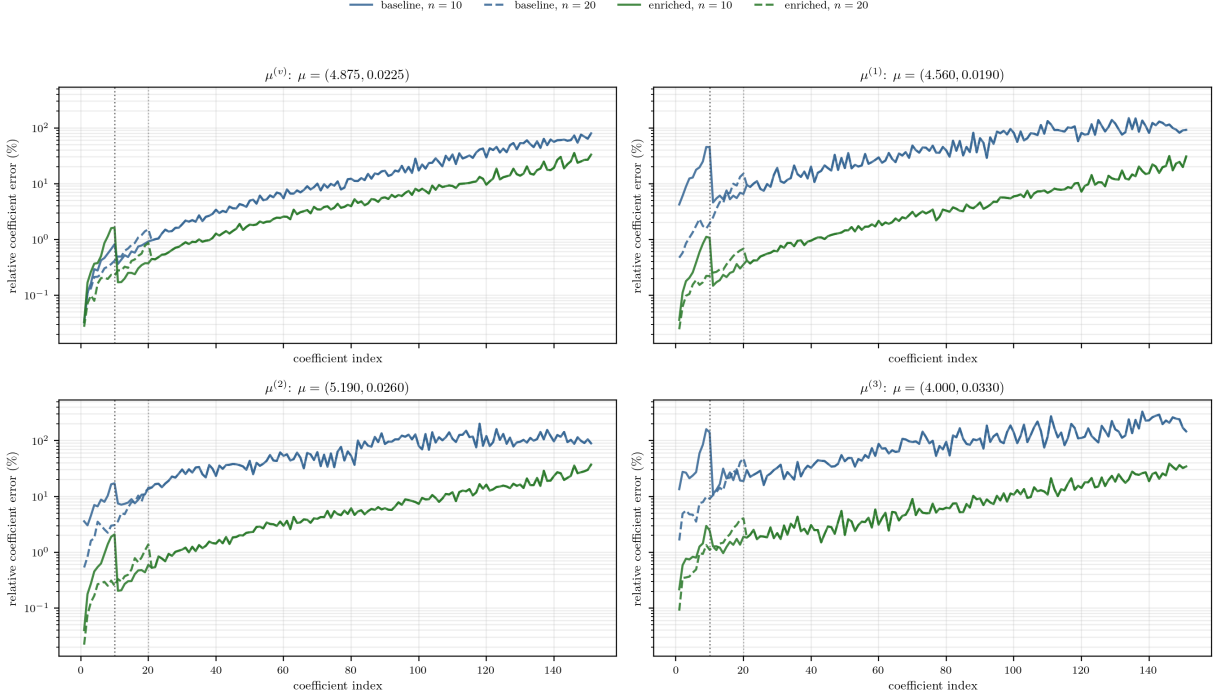


Figure 18: Case-2 per-coefficient HPROM errors before and after coefficient-data enrichment. Dotted vertical lines mark coordinates 10 and 20. The $n = 10$ and $n = 20$ variants use the same master-map tail, so their enriched curves coincide after coefficient 20; their difference is confined to coordinates recovered by the residual solve.

Figure 18 gives the mechanism behind the aggregate results. In the baseline, the injected tail is inaccurate at most secondary coordinates, especially away from the training grid. Enrichment substantially lowers those tail errors. The sampled residual solve then needs only to correct the leading coordinates, which is why even Case 2 with $n = 10$ returns to the linear-HPROM state-accuracy scale.

11 Conclusions

The PROM and HPROM studies support the following conclusions.

1. The fixed MLSPG-sensitive linear PROM provides a reproducible coefficient teacher and state-accuracy reference for this benchmark.
2. With only nine training trajectories, direct parameter-time prediction of all 151 coefficients is the limiting task. This affects both the POD-NN-ROM and the Case 2 injected tail.
3. Solving more primary coordinates in Case 2 can help, but the matched-rule HPROM sweep is non-monotone at small n and does not reach the linear-HPROM reference by $n = 50$. With a fixed $n = 10$ rule, imposed tail error changes the primary coordinates recovered by the residual solve; tail accuracy is therefore essential.
4. PROM data enrichment changes the picture substantially. Keeping the architecture and basis fixed, the 9+36 PROM trajectory set reduces validation and online coefficient errors and brings the learned state errors close to the linear PROM level.

5. The baseline HPROM with 2% parameter–time stratified snapshots preserves the main PROM conclusion for Case 1, Case 3, and PROM–POD–AE: these learned intrusive models remain close to the linear HPROM reference at in-domain points.
6. Case 2 is the residual-sampling stress test. With nine coefficient trajectories, $n = 10$ is highly sensitive to the injected secondary tail. The fixed-rule diagnostic demonstrates that this sensitivity is residual feedback, not solely reconstruction error, while the matched-rule sweep shows improvement only once a sufficiently large primary block is solved.
7. HPROM coefficient-data enrichment transfers the PROM-level gain without changing the linear HPROM reference or adding HDM training snapshots. The $9 + 36$ campaign reduces Case–2 $n = 10$ from 2.641% to 0.468% in-domain and from 13.098% to 0.921% at extrapolation, placing it on the same practical accuracy scale as the linear HPROM.
8. The same enrichment effect is present without an online residual solve. Both direct maps improve sharply, particularly at extrapolation, confirming that coefficient-data coverage is a primary limitation. These maps do not quantify ECM error, however, so they complement rather than replace the intrusive HPROM comparisons.